

Data Preprocessing and EDA

Course Code	Course Title	Credits	Type (T+P+Pj)
CUGG1004	Data Preprocessing and EDA	4	2+2+0

Course Objectives:

This course aims to provide students with:

- To introduce the fundamentals of data preprocessing techniques for different data types.
- To develop skills in handling missing data, outliers, and data scaling for machine learning models.
- To explore advanced feature engineering and dimensionality reduction techniques.
- To enhance data visualization skills for effective data interpretation and decision-making.
- To provide hands-on experience with real-world datasets, preprocessing pipelines, and automated data handling tools.

Course Outcomes:

After the completion of the course students will be able to:

- CO1: Apply data preprocessing techniques to clean and prepare raw datasets for analysis.
- CO2: Analyze data features and implement feature engineering techniques to improve model performance.
- CO3: Develop data visualization techniques to uncover patterns and insights from data.
- CO4: Evaluate preprocessing pipelines for diverse datasets using automation tools.
- CO5: Design workflows for handling time-series, text, and imbalanced datasets effectively

Course Syllabus:

Module 1: Preprocessing Basics

Theory

- Data Preprocessing Basics: raw data formats, data cleaning techniques
- Handling Missing & Categorical Data: imputation methods, one-hot and label encoding
- Outliers & Scaling: outlier detection and removal, scaling and normalization techniques

Practice

- Experiment 1.1: Write a program to load datasets (CSV/JSON) and identify missing values
- Experiment 1.2: Implement imputation techniques (mean, median, mode) for handling missing data
- Experiment 1.3: Develop a program to encode categorical variables using one-hot and label encoding
- Experiment 1.4: Write a program to detect and remove outliers using Z-score and IQR methods

- Experiment 1.5: Implement data scaling techniques such as Min-Max normalization and standardization

Module 2: Feature Engineering

Theory

- Feature Selection: correlation methods, statistical tests
- Dimensionality Reduction: PCA, t-SNE techniques
- Feature Creation: interaction terms, polynomial features, importance evaluation

Practice

- Experiment 2.1: Write a program to perform feature selection using correlation analysis
- Experiment 2.2: Implement statistical methods for feature selection
- Experiment 2.3: Develop a program to apply Principal Component Analysis (PCA)
- Experiment 2.4: Write a program to perform dimensionality reduction using t-SNE
- Experiment 2.5: Implement creation of polynomial features and interaction terms

Module 3: Data Visualization

Theory

- Visualization Tools: Matplotlib, Seaborn libraries
- Plot Types: bar, line, scatter, histograms, heatmaps
- Interactive & Insights: Plotly visualizations, correlation analysis, decision-making

Practice

- Experiment 3.1: Write a program to visualize data using Matplotlib (line, bar, scatter plots)
- Experiment 3.2: Implement data visualization using Seaborn (histograms, box plots)
- Experiment 3.3: Develop a program to generate heatmaps for correlation analysis
- Experiment 3.4: Write a program to create interactive visualizations using Plotly
- Experiment 3.5: Implement visualizations to support data-driven decision-making

Module 4: Case Studies & Tools

Theory

- Pipelines & Tools: Scikit-learn pipelines, AutoML introduction
- Data Analysis: real-world cleaning, open dataset exploration
- Projects: group-based exploratory data analysis tasks

Practice

- Experiment 4.1: Write a program to build preprocessing pipelines using Scikit-learn
- Experiment 4.2: Implement data cleaning on a real-world dataset
- Experiment 4.3: Develop a program to analyze datasets from open sources (Kaggle/UCI)
- Experiment 4.4: Write a program to demonstrate basic AutoML tools
- Experiment 4.5: Perform exploratory data analysis as a mini group project

Module 5: Advanced Data Handling

Theory

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